

# **DAYANANDA SAGAR COLLEGE OF ENGINEERING**

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning**



## **INTERNSHIP REPORT**

**ON**

### **“Data Analyst”**

*Submitted in partial fulfillment for the award of degree(20AI8ICINT)*

## **BACHELOR OF ENGINEERING IN**

**Department of Artificial Intelligence and Machine Learning**

*Submitted by:*

**Rohan D - 1DS20AI043**

Conducted at



**Peoplebox.ai**

**Department of Artificial Intelligence and Machine Learning  
Dayananda Sagar College of Engineering Bangalore-78**

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



## CERTIFICATE

This is to certify that the Internship titled “**Data Analyst**” carried out by **Mr ROHAN D [1DS20AI043]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

\_\_\_\_\_  
**Signature of Reporting  
Head**  
**Chaitanya Patil**  
MDM Team  
Lead,  
Peoplebox.ai

\_\_\_\_\_  
**Signature of Internship  
Coordinator**  
**Prof. Kavya D N**  
Dept. of AI & ML  
DSCE

\_\_\_\_\_  
**Signature of HoD**  
**Dr. Vindhya P Malagi**  
Dept. of AI & ML  
DSCE

### External Viva:

Name of the Examiner

1) \_\_\_\_\_

2) \_\_\_\_\_

Signature with Date

\_\_\_\_\_

\_\_\_\_\_

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning



## DECLARATION

I, **Rohan D**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, at **Peoplebox.ai**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI043

Place: Bangalore

NAME: Rohan D

# OFFER LETTER



17th January 2025

Dear Rohan Divakar,

With reference to your discussion with us, We, Culturegrade Technologies Pvt Ltd ("Peoplebox", "Company"), are pleased to make you an offer in our organization as **Data Analyst Intern**. This offer takes effect from your date of joining on no later than **20th January 2025**

## Stipend

Your stipend during the internship would be **INR 10,000/- per month** and will be due and paid to you at the end of each calendar month, or on normal pay periods which the Company decides upon.

## Internship Period

As per organization policy, the internship period applicable to you shall be three (3) months. During internship, the period of notice required for resignation is 15 days on either side. After the internship, based upon the company's requirements and your performance the company will consider you for a full-time position.

## Policies

1. You will abide by the rules and regulations of the Company as may be in force from time to time and if any violation made would be subjected to the Disciplinary action.
2. This letter of offer is based on the information furnished in your application for employment and during the interviews you had with us. If, at any time in future, it comes to light that any of this information is incorrect or any relevant information has been withheld, then your employment is liable to be terminated without notice.
3. The work location is based out of Bangalore.

## Reference Checks

Your appointment is subject to satisfactory reference checks and clearance from any secrecy / service agreements that you may have executed, which could have a bearing on your working with us.

## Confidentiality Non Disclosure & Intellectual Property Rights

As a condition to this employment, you are required to sign the company's standard confidentiality, Non Disclosure and Intellectual Property Rights agreement at the time of your joining.

The company retains ownership of the intellectual property rights (relating to discoveries, developments, improvements, processes, formulae, algorithms, works of authorship including software programs, user

# **COMPLETION CERTIFICATE**

## ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Chaitanya Patil, MDM Team Lead, Peoplebox.ai** for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

**ROHAN D [1DS20AI043]**

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# CHAPTER 1

## INTRODUCTION

### **BUILDING TARA: A DATA-DRIVEN APPROACH TO AI TALENT ACQUISITION**

At Peoplebox.ai, we are at the forefront of building AI-powered solutions to transform the way organizations hire and manage talent. One of our most ambitious projects is *Tara* — a next-generation AI talent acquisition agent designed to intelligently screen candidates, match them with relevant roles, and assist hiring teams in making faster, data-informed decisions.

As a Data Analyst intern on the Tara product, my primary responsibility is to collect, structure, and extract high-quality data from resumes. This data forms the foundation for training and fine-tuning the machine learning models that power Tara. The goal is to help the AI agent learn how to understand and evaluate diverse candidate profiles, skill sets, and experience levels, mimicking the decision-making capabilities of a seasoned recruiter.

The process involves rigorous screening of resumes to identify and label critical information such as educational qualifications, work experience, technical and soft skills, certifications, and more. By ensuring the data is clean, consistent, and accurately labeled, we significantly enhance the model's ability to learn, generalize, and make reliable predictions.

#### **Key Benefits and Impact of My Work:**

- 1. Faster and Smarter Hiring**

With well-structured data, Tara can automate the initial screening process, saving recruiters hours of manual effort and helping companies fill roles faster.

- 2. Scalable Talent Intelligence**

The data pipeline we build supports the analysis of large volumes of resumes, making the AI agent capable of handling enterprise-level hiring demands.

- 3. Empowering HR Teams**

Through data-driven insights, recruiters and HR managers can make better decisions about candidate selection, team structuring, and long-term workforce planning.

In essence, my role is a crucial part of the AI development lifecycle at Peoplebox.ai. By transforming raw, unstructured resume data into high-quality training inputs, I contribute directly to building Tara into a reliable, intelligent, and impactful AI tool that is shaping the future of talent acquisition.

## CHAPTER 2

### DESCRIPTION OF ORGANIZATION



#### Peoplebox.Ai

##### Introduction

**Peoplebox.ai** was founded in 2019 with the mission of transforming how companies manage performance, align goals, and drive business outcomes using AI. Its vision is to help organizations build high-performing teams by simplifying and automating performance management and talent acquisition through intelligent tools.

From its inception, Peoplebox.ai has focused on leveraging cutting-edge technologies such as AI and machine learning to solve challenges in human resource management. Its flagship platform integrates performance reviews, OKRs (Objectives and Key Results), 1:1 meetings, and employee engagement tools in one seamless experience. With its data-driven insights, Peoplebox enables leadership teams to make strategic decisions and build a culture of accountability and growth.

In 2025, Peoplebox introduced **Tara & Nova**, a generative AI-powered talent acquisition & talent management agent designed to enhance the recruitment process. Tara uses machine learning to extract critical data from resumes, screen candidates, and help recruiters make faster, smarter hiring decisions—drastically improving time-to-hire and candidate quality.

Peoplebox.ai has achieved several milestones in recent years. The platform has been adopted by over 500 fast-scaling companies, including Razorpay, Redbus, Whatfix, and Exotel. In 2022, the company was selected by Y Combinator, one of the world's most prestigious startup accelerators, for its Summer batch. This recognition enabled Peoplebox to gain global visibility and expand its impact across industries.

To enable seamless team collaboration, Peoplebox integrates natively with popular tools such as Slack, Microsoft Teams, Jira, Asana, and more. These integrations allow users to track goals, give feedback, and conduct reviews without leaving their daily workspace, increasing efficiency and adoption.

With a strong commitment to innovation and simplicity, Peoplebox is also investing in future-ready solutions like predictive analytics in HR, AI-led workforce planning, and deeper automation of performance processes. These developments are aimed at empowering leaders to proactively manage talent and build adaptable, resilient organizations.

## **Founders**

### **Abhinav Chugh**

Abhinav Chugh, co-founder and CEO of Peoplebox.ai, is a seasoned entrepreneur with a passion for building products that enhance organizational efficiency. An alumnus of the Indian Institute of Technology (IIT) Delhi, Abhinav has over 16 years of experience in the tech industry, having worked with startups and multinational corporations alike. His vision for Peoplebox.ai stems from a deep understanding of the challenges organizations face in aligning people strategies with business objectives.

### **Alagu Muthuraman**

Co-founder Alagu Muthuraman brings a wealth of experience in product development and engineering. With a background in building scalable tech solutions, Alagu has been instrumental in shaping the technical architecture of Peoplebox.ai. His expertise ensures that the platform remains robust, user-friendly, and adaptable to the evolving needs of modern organizations.

## **Conclusion**

Peoplebox.ai stands at the forefront of HR technology, offering innovative solutions that address the complexities of talent management and organizational alignment. Through its AI-driven platform, the company empowers businesses to make informed decisions, foster employee engagement, and achieve strategic objectives with precision.

As organizations navigate the challenges of the modern workplace, tools like Peoplebox.ai become indispensable in driving performance and sustaining growth. With a commitment to continuous innovation and a focus on delivering tangible results, Peoplebox.ai is poised to redefine the future of work.

## ABOUT THE COMPANY

### About Peoplebox.ai

Peoplebox.ai is a Bengaluru-based HR-tech company dedicated to building AI-powered solutions for performance management and talent operations. Their flagship platform enables organizations to manage OKRs, performance reviews, one-on-ones, and employee engagement, all in one place. Trusted by over 500 high-growth companies across 50+ countries, Peoplebox's solutions are designed to help leaders drive alignment, accountability, and transparency within their teams.

With deep integrations across platforms like Slack, Google Workspace, Microsoft Teams, and various HRMS tools, Peoplebox ensures seamless workflow adoption. Their AI assistant, **Tara**, is built to support recruiters and HR teams by streamlining hiring, tracking talent metrics, and offering data-driven insights for better people decisions.

Backed by investors including Y Combinator and Kae Capital, Peoplebox continues to innovate at the intersection of AI and HR operations, enabling companies to build performance-driven cultures that scale.

## VISION

To become the world's most trusted AI-powered People Operating System, helping organizations drive performance, alignment, and growth by transforming the way teams set goals, give feedback, and make talent decisions.

## MISSION

To empower organizations with smart, integrated, and user-friendly solutions that align people operations with business outcomes—by simplifying performance management, automating HR workflows, and enabling data-driven decisions through AI and deep integrations.

## CHAPTER 3

### SCOPE OF WORK

During my internship at **Peoplebox.ai** as a **Data Analyst**, I was responsible for building and automating a complete, production-grade data pipeline that enabled the fine-tuning of GPT-4o, powering Tara — Peoplebox's AI talent acquisition agent. The core objective was to transform raw resumes into structured, validated, and explainable datasets suitable for model training. This pipeline consisted of two major components:

#### Step 1: Resume Parsing & Structuring with LLMs

##### Objective:

Automate the end-to-end extraction and transformation of resumes into structured formats, enabling large-scale data generation for machine learning.

##### Methodology & Tools:

- Designed and developed a Python-based resume parser from scratch.
- Used `fitz` (PyMuPDF) for layout-aware PDF parsing and `pytesseract` with `PIL` for OCR of scanned/image-based resumes.
- Extracted raw text was passed to **Gemini 2.0 Flash** via API, which was prompted to return structured JSON containing:
  - Candidate name
  - Company names
  - Job titles
  - Employment timelines
  - Detailed experience data
- A second call to Gemini was made using custom prompts and a hierarchical job role list to assign:
  - High Match and Low Match job roles
  - Associated IC (Individual Contributor) levels

##### Data Validation & Storage:

- Implemented robust validation and error handling logic for role accuracy and deduplication.
- Stored results in **MongoDB** and generated a CSV for team review.

##### Impact:

- Enabled scalable, automated generation of training data.
- Significantly reduced manual resume processing efforts by over 60%.
- Created standardized, reviewable datasets that formed the training foundation for Tara.

## Step 2: Generating Justification Thoughts for Fine-Tuning

### Objective:

Enhance dataset interpretability by generating reasoning (“thoughts”) for each prediction made by the AI — to be used for fine-tuning and auditing GPT-4o.

### Methodology & Tools:

- Built a second automated system to process CSV records from Step 1.
- For each row, used Gemini 2.0 Flash to generate:
  - **Technical Skills Thoughts:** Why the assigned skills match the experience.
  - **Job Role – High Thoughts:** Justification for the primary assigned job role and IC level.
  - **Job Role – Low Thoughts:** If applicable, explanation for the secondary role and level.
- Structured the outputs in both **CSV** (for manual QA) and **JSONL** (for direct use in GPT-4o fine-tuning).
- Ensured responses were within a 100-word limit and followed strict formatting and validation rules.

### Impact:

- Created model-trainable data with explainability, aiding in transparency and review.
- Directly contributed to the fine-tuning of the GPT-4o model used by Tara.
- Enabled the AI to not only recommend roles and skills but also explain its reasoning — critical for building trust with hiring teams.

## Conclusion

Through the automation of both resume parsing and justification generation, I created a two-step data pipeline that formed the backbone of GPT-4o’s training loop at Peoplebox.ai. This pipeline converted thousands of unstructured resumes into structured, labeled, and explainable datasets — accelerating model development and enhancing Tara’s precision in talent recommendations. My contributions empowered the Peoplebox team to train their models with higher-quality data, faster iteration cycles, and human-auditable logic, playing a key role in scaling the AI recruitment engine.

## CHAPTER 4

### LEARNING OBJECTIVES

#### 1. Mastery of Machine Learning and LLM-Based Systems

Develop a strong grasp of applied machine learning and large language model (LLM) integration in real-world HR-tech systems. The objective was to gain hands-on experience in fine-tuning advanced language models such as **GPT-4o** and **Gemini 2.0 Flash** using structured datasets derived from raw resume data. Through active contribution to model development and validation processes, the goal was to master the creation and deployment of intelligent systems for resume parsing, role classification, and talent recommendation.

#### 2. Advanced Data Engineering and Automation Skills

The internship focused on developing expertise in designing and automating large-scale data pipelines. Interns were tasked with handling noisy, unstructured resume data and transforming it into structured formats. Skills in data validation, error handling, CSV/JSONL formatting, and scalable file processing were emphasized. I independently build and automate ETL workflows that integrated with LLM APIs, enabling the seamless generation of training datasets for downstream ML tasks.

#### 3. Understanding of Talent Intelligence and HR-Tech Systems

Gain comprehensive knowledge of how data and AI power talent acquisition systems in modern HR-tech environments. Interns were encouraged to study the lifecycle of recruitment pipelines, job role taxonomies, IC level hierarchies, and skills mapping. This involved understanding how resume data translates into actionable HR insights and contributes to tools like **Tara**, Peoplebox's AI talent agent. Through practical exposure and collaboration with product and data teams, the aim was to develop domain awareness critical for designing intelligent talent systems.

#### 4. Proficiency in End-to-End AI System Development

Achieve proficiency in building full-stack data and AI components — from raw data ingestion to model-ready outputs and downstream fine-tuning. Interns were expected to independently design, implement, and test modular systems that automate data extraction, LLM prompting, and formatted output generation. This included handling both technical and business logic layers, ensuring consistency and correctness across the pipeline. Emphasis was placed on integrating components like resume parsers, classification engines, explainability modules, and MongoDB storage into a cohesive, scalable system.

## 5. Application of Generative AI in Talent Intelligence

Understand and apply the capabilities of generative AI to revolutionize talent acquisition. I explored how LLMs like **Gemini** and **GPT-4o** can be fine-tuned to perform tasks such as resume analysis, role justification, IC level inference, and skills explanation. By designing custom prompt templates, analyzing model outputs, and preparing JSONL datasets for model training, interns contributed directly to the evolution of AI agents in recruitment. The objective was to demonstrate thought leadership in applying AI to solve complex problems in the HR-tech industry.

## CHAPTER 5

# CHALLENGES AND SOLUTION

### 1. Resume Parsing and Preprocessing Pipeline

#### Challenges:

- **Unstructured and Diverse Resume Formats:**

Resumes varied widely in format, language, layout, and structure (scanned images, text-based PDFs, etc.), making uniform data extraction highly challenging.

- **OCR Accuracy and Noise Removal:**

Extracting reliable text data from scanned/image-based resumes using OCR (pytesseract) often resulted in noise, misreads, and inconsistent formatting, affecting the quality of downstream tasks.

- **Information Extraction and Entity Identification:**

Mapping extracted text to relevant fields like company names, job roles, timelines, and skills required careful tokenization, pattern matching, and standardization due to inconsistent phrasing across resumes.

- **LLM Prompt Design and Rate Limiting:**

Designing concise yet informative prompts for Gemini 2.0 Flash to return structured output was a major challenge. Ensuring fast, accurate responses within token limits while avoiding hallucinations or incomplete results demanded prompt engineering expertise.

- **Data Validation and Storage:**

Automating QA for extracted outputs, preventing duplicates, and designing consistent formats for MongoDB and CSV/JSONL exports required extensive testing, error handling, and logging mechanisms.

#### Solution:

To address these challenges, I designed and developed a complete **resume parsing pipeline** from scratch. Using fitz (PyMuPDF) for text extraction and pytesseract for OCR, I handled all resume types and routed the parsed content through **Gemini 2.0 Flash** via API to generate structured data (e.g., name, job titles, companies, timelines, experience summaries). I also implemented:

- Prompt-based role classification using a hierarchical job taxonomy.
- Validation logic for incomplete/invalid fields.
- Storage integration with MongoDB and CSV for downstream review.

This resulted in a highly scalable, modular data pipeline that served as **Step 1 of the GPT-4o fine-tuning workflow** for Tara.

## 2. Thought Generation and Data Structuring for Model Fine-Tuning

### Challenges:

- **Contextual Justification Generation:**

Generating concise, role-specific justifications for assigned skills and job roles required advanced LLM reasoning. Gemini needed to interpret nuanced candidate experience to explain decisions in a human-readable form.

- **Prompt Crafting for High-Accuracy Responses:**

Ensuring that Gemini's responses were consistent, under 100 words, and followed a strict output format involved trial-and-error tuning of prompt templates across multiple data points.

- **Handling Incomplete Role Assignments:**

Many resumes lacked either a Low Match role or IC level, requiring logic to bypass irrelevant prompts and avoid unnecessary API calls.

- **Bulk Processing with Error Handling:**

Processing large datasets required batching and tracking each response for errors or inconsistencies. Failed LLM calls had to be retried, logged, or flagged for manual review.

- **Structured Output for Training (JSONL Format):**

Creating training-ready files in **JSONL format** (used for fine-tuning GPT-4o) with messages, role annotations, and assistant responses required careful structuring and consistency.

### Solution:

To meet these challenges, I built the **Thought Generator pipeline** which:

- Generated reasoning for each resume's assigned roles, IC levels, and technical skills using Gemini 2.0 Flash via well-structured prompts.
- Outputted both CSVs (for human review) and JSONL files (for fine-tuning).
- Included clean formatting of multi-level role-thought mappings, consistent schemas, and fallback logic for missing fields.

This allowed Peoplebox.ai to create a transparent, explainable dataset that contributed directly to **fine-tuning GPT-4o**, improving Tara's reasoning and recommendation quality.

### 3. Building a Scalable End-to-End AI Data Pipeline

#### Challenges:

- **Automation with Flexibility:**

Balancing automation with flexibility (e.g., handling different batch sizes, formats, and custom prompt sets) while ensuring ease of use for future contributors.

- **Cross-Team Collaboration for QA:**

Building CSVs that the data and ML teams could easily review, verify, and correct without disrupting downstream processes.

- **Data Drift and Continuous Iteration:**

New resume formats and edge cases required frequent updates to parsing logic, prompt design, and formatting standards to maintain consistency.

#### Solution:

I modularized both pipeline stages, implemented CLI integration for dynamic input/output handling, and set up configurable batch limits for safe bulk processing. The system was fully self-contained, maintainable, and production-ready — allowing the team to scale fine-tuning with minimal manual intervention.

#### Conclusion

These challenges were crucial to building Tara's foundation as an intelligent AI recruitment agent. Through structured resume parsing, smart role and skill classification, and explainable AI dataset generation, my work ensured the Peoplebox ML team could **fine-tune GPT-4o with clean, interpretable, and high-quality training data**. My end-to-end pipeline not only enhanced the accuracy and explainability of the AI but also significantly reduced turnaround time and manual data wrangling, positioning Tara for large-scale deployment.

# HARDWARE AND SOFTWARE REQUIREMENTS

## Software Requirements:

- Operating system: macOS Sonoma 14.x (Apple Silicon)
- Python: 3.12.0
- Pandas: 2.2.0
- PyMuPDF (fitz): for structured text extraction from PDFs
- Pytesseract: for OCR on scanned/image-based resumes
- google.generativeai: for Gemini 2.0 Flash API integration
- Anaconda Distribution: 2023.09
- Cursor IDE Community Edition: 2024.1

## Hardware Requirements:

- Processor: Apple M3 Pro
- RAM: 18 GB unified memory
- Display: 14" Display
- Input Devices: Built-in Keyboard and Trackpad (Mouse optional)
- Internet: Required (Wi-Fi or LAN, Fiber Optic for stable Gemini API access)
- Storage: 512 GB SSD (with sufficient space for logs, CSVs, and JSONL outputs)

## CHAPTER 6

### ACHIEVEMENTS AND CONTRIBUTIONS

During my internship at **Peoplebox.ai** as a **Data Analyst**, I worked on a high-impact project that involved developing and deploying an end-to-end automated data pipeline used for fine-tuning GPT-4o — the core engine powering *Tara*, Peoplebox's AI-driven talent acquisition agent. The pipeline was divided into two main phases: **Resume Parsing and Structuring** and **Thought Generation and Justification**.

#### 1. Resume Parsing and Structuring (Step 1 – resume\_parser.py)

##### Data Preprocessing and Analysis:

The first challenge involved building a system to extract structured information from resumes that were received in varied formats including text-based PDFs and scanned documents. This stage formed the foundation of the AI training dataset and had to be accurate, scalable, and robust against formatting inconsistencies.

To address this, I created a parsing engine from scratch using:

- fitz (PyMuPDF) for parsing text and layout from PDFs
- pytesseract for OCR-based parsing of image-based or scanned resumes
- pandas for processing and cleaning extracted data

From each resume, key information was extracted such as:

- Candidate Name
- Company Names
- Job Titles
- Timeline Patterns (Start Date, End Date, Duration)
- Detailed Experience Text
- Technical Skills

After cleaning and structuring the raw content, the data was passed to **Gemini 2.0 Flash** via API, which generated structured JSON outputs containing:

- High Match Job Role and IC Level
- Low Match Job Role and IC Level (if applicable)
- Parsed Technical Skills

All responses were validated, deduplicated, and stored in **MongoDB** for review and future querying. Additional exports were generated in **CSV** for team review and **JSONL** format for model fine-tuning.

## Resume Parsing Workflow

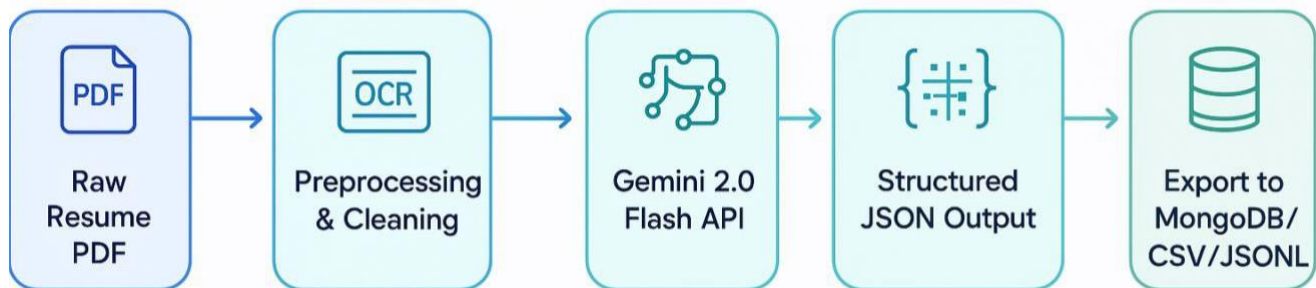


Fig 6.1: Resume Parsing Workflow

### Impact:

- Reduced resume processing time from **multiple weeks to just a few hours**.
- Enabled the creation of **high-quality structured training datasets** for machine learning and fine-tuning tasks.
- Significantly reduced manual labor, previously required for entity extraction and formatting.
- Allowed the team to scale dataset generation by **5–10x** without increasing headcount.

## 2. Thought Generation and Justification (Step 2 – generate\_thoughts.py)

### Modeling and Implementation:

Once the resume data was structured, the next step was to **enrich the dataset with reasoning** — short, AI-generated explanations for the job roles, IC levels, and technical skills assigned to each candidate. This was crucial for improving **transparency and explainability** in the AI model's recommendations.

To accomplish this, I developed a second pipeline that:

- Accepted parsed resume content from the first step
- Created tailored prompts for each candidate to explain:
  - Why the **technical skills** matched the experience
  - Why the **primary job role and IC level** were appropriate
  - Why the **secondary role (if applicable)** fit the experience

These prompts were passed to **Gemini 2.0 Flash**, and the returned thoughts were:

- Attached to their corresponding candidate record
- Stored in **CSV** format for manual review
- Exported to **JSONL** files to be used directly for GPT-4o fine-tuning

#### **Impact:**

- Enabled generation of **hundreds of explainable records within minutes**, replacing the need for human-written annotations.
- Created training-ready chat-style JSONL data in the format:

**System → User → Assistant**

- Improved **explainability** of Tara's model outputs, allowing HR professionals to understand and trust AI-driven decisions.

### **3. Full Pipeline Integration and Automation**

After building both components, I unified them into an **end-to-end, production-ready pipeline** that could:

- Ingest resumes in batch mode
- Parse and extract structured data
- Classify job roles and skills
- Generate LLM-based explanations
- Export ready-to-train files in JSONL and CSV

Features included:

- CLI support for single and batch CSV input
- Automatic folder creation for outputs
- API rate handling and error logging
- Optional row limits for testing and safe deployments

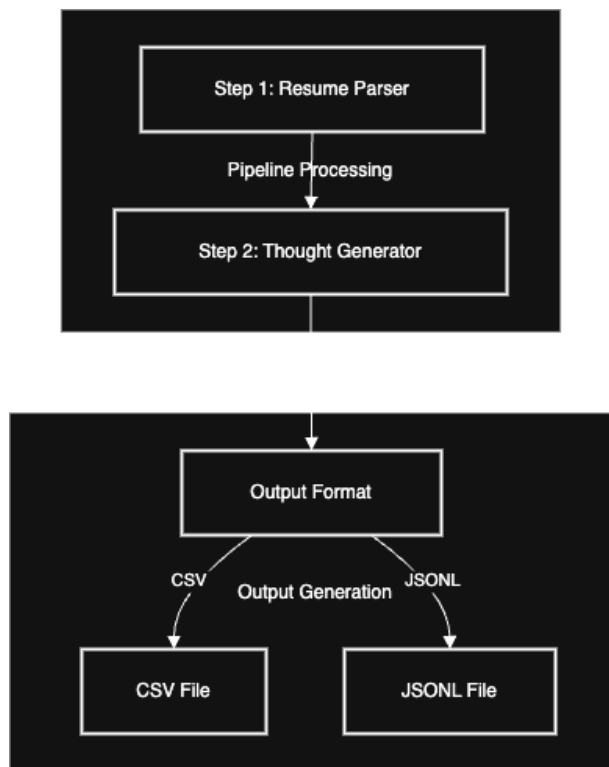


Fig 6.2: Output Files for Review & Fine-Tuning

#### Impact:

- Transformed a previously manual, error-prone, multi-week process into a **streamlined pipeline completed in a few hours**.
- Created a solution that can scale to process **thousands of resumes automatically**.
- The pipeline became a **core component in the training and validation of Tara**, Peoplebox's production-grade AI.

#### 4. Recognition and Lasting Impact

- My work was **personally acknowledged and appreciated** by the **co-founders and senior leadership** at Peoplebox.ai.
- The systems I built have left an **everlasting technical footprint** in the company — continuing to support Tara's evolution beyond the scope of my internship.
- Contributed directly to building a **smarter, explainable, and scalable AI hiring assistant**, deployed across real-world HR operations.

## **Final Contribution Summary**

By independently designing, building, and automating two core systems — Resume Parsing and Thought Generation — I enabled the Peoplebox team to:

- Accelerate GPT-4o training cycles
- Improve the explainability and trustworthiness of AI recommendations
- Save weeks of manual effort across technical and non-technical teams

What once took the team weeks, now takes hours.

And what once required manual effort, now runs at scale — with clarity, consistency, and intelligence.

## CHAPTER 7

### REFLECTION AND ANALYSIS

During the course of my internship at **Peoplebox.ai**, I gained extensive practical exposure to the world of **data engineering, applied machine learning, and LLM integration** — all within the real-world context of building an AI talent acquisition platform. This project significantly expanded my understanding of how unstructured information can be systematically converted into structured, explainable data pipelines that serve advanced AI applications like **GPT-4o**.

One of the most transformative learning experiences was mastering data handling techniques across a wide variety of resume formats. Parsing and processing resumes that came in both structured and scanned formats challenged me to explore technologies. Through this, I developed a strong understanding of text extraction, layout preservation, and OCR optimization. I also learned how to handle inconsistencies and missing information, ensuring data quality through robust preprocessing and validation steps.

My involvement in **end-to-end pipeline automation** helped me appreciate the critical role of modular design, logging, error handling, and flexibility in real-world deployments. Building the system to convert resume content into structured fields, feeding it to Gemini 2.0 Flash, and formatting the output in CSV and JSONL formats taught me how multiple layers of a data pipeline interconnect and how important clean, well-structured inputs are for training large language models.

Working with **LLMs like Gemini 2.0 Flash** introduced me to the nuances of **prompt engineering**, contextual prompting, and rate-efficient API integration. Designing prompts to elicit consistent, short, and accurate "thoughts" required creativity and precision. I gained a new appreciation for how natural language interfaces can be adapted for structured outputs in production ML pipelines.

This project also sharpened my skills in **data visualization and explainability**, especially while building justification-generating modules. Producing explainable outputs for job roles, skills, and IC levels forced me to think deeply about what makes AI interpretable and trustworthy. Generating these explanations and reviewing them with the team helped reinforce the value of transparency in AI systems.

Perhaps most importantly, this internship offered a deeper understanding of how AI and data science are **applied meaningfully in the HR-tech domain**. Seeing how my scripts directly contributed to Tara — the AI assistant — and helped the company train GPT-4o more effectively, gave me a strong sense of purpose. The real-world impact of my code went beyond theoretical model building — it empowered a tool that may soon help thousands of companies recruit smarter, faster, and more fairly.

The feedback and appreciation I received from the co-founders and senior leadership at Peoplebox further validated the importance and effectiveness of my work. Knowing that my pipeline will continue to power Tara's evolution beyond my internship gave me immense confidence and reinforced my passion for scalable, impactful data systems.

In conclusion, this internship has been a turning point in my professional development. It provided hands-on experience with data pipelines, LLMs, prompt engineering, and ML dataset preparation — all within the framework of a real, deployed product. It helped me bridge the gap between academic learning and production-level implementation, preparing me for a future in AI, machine learning, and intelligent system design.

## CHAPTER 8

### RECOMMENDATIONS

#### 1. Introduce Structured Internship Frameworks

To further enhance the internship experience at Peoplebox.ai, implementing a more **structured program** with clearly defined **learning objectives**, **project scopes**, and **milestone-based evaluations** would provide interns with a more focused roadmap. A balance between technical training (e.g., APIs, LLM integration, data engineering best practices) and project-based execution can greatly help interns align their day-to-day work with broader company goals. Regular guidance sessions or sprint reviews could ensure steady progress and help measure skill development over the course of the internship.

#### 2. Strengthen Academic Collaboration

Partnering with academic institutions through **capstone projects**, **guest lectures**, or **collaborative research initiatives** could open up new avenues for talent discovery and innovation. Inviting students to contribute to ongoing work in LLM fine-tuning, prompt engineering, and AI-driven recruitment platforms (like Tara) would foster a real-world connection between academic theory and industrial application. Such programs could also allow students to explore emerging technologies like GPT-4o or Gemini within a practical business environment.

#### 3. Expand Mentorship and Peer Review Systems

While the Peoplebox team was incredibly supportive, introducing a **formal mentorship structure** where each intern is paired with a dedicated mentor could help maximize learning outcomes. Mentors could assist in reviewing code, brainstorming LLM prompt strategies, or even helping interns understand industry-specific challenges in the HR tech domain. Structured weekly or biweekly 1:1s, coupled with access to cross-functional feedback, would create a more well-rounded developmental environment.

#### 4. Encourage Innovation through Internal AI Labs or Hackathons

Given the AI-centric nature of Peoplebox.ai, creating space for interns to **experiment with new ideas** — such as resume anonymization, job-role inference optimization, or explainability tools — would encourage innovation and a sense of ownership. Hosting **mini hackathons**, **innovation sprints**, or “**build week**” **challenges** could empower interns to think beyond their assigned tasks and explore solutions that might benefit the company in unexpected ways.

## **5. Facilitate Cross-Team Interaction and Networking**

Interns benefit greatly from exposure to different domains. Organizing casual networking sessions, tech talks, and knowledge-sharing forums across product, ML, engineering, and design teams would enrich the learning experience. Providing platforms (like Slack channels or monthly demo days) where interns can present their work and gain feedback from senior team members and peers will also foster a sense of community and long-term professional connection.

## CHAPTER 9

### CONCLUSION

In conclusion, my internship at Peoplebox.ai focused on solving a critical challenge in the development of AI-powered talent acquisition systems — the need for structured, high-quality, and explainable data for fine-tuning large language models (LLMs). By independently designing and developing a two-stage data pipeline, I contributed directly to the creation of Tara, the company's AI agent designed to revolutionize recruitment through intelligent decision-making.

The first stage of the pipeline focused on resume parsing and structuring to process diverse resume formats. By leveraging the capabilities of Gemini 2.0 Flash through API integration, I was able to extract meaningful job role and skill mappings from unstructured candidate experiences. This process was automated and standardized, ensuring that resume data could be consistently transformed into model-ready input. The second stage focused on generating explainable insights, or “thoughts,” for each candidate's technical skill set and role assignments. These justifications were generated via prompt-engineered LLM calls and formatted in both CSV and JSONL formats, aligning directly with GPT-4o fine-tuning requirements. The outcome was a transparent, explainable dataset that improved both training quality and trust in model outputs.

The successful implementation of this pipeline significantly reduced manual effort, transforming a process that previously took weeks into one that now takes just a few hours. More importantly, it provided the ML team at Peoplebox with a scalable foundation for expanding their AI-driven capabilities — from resume understanding to intelligent role recommendations.

Beyond technical achievements, this project served as a practical example of how modern LLMs and deep learning can be effectively integrated into enterprise workflows, especially in domains like HR-tech where explainability, consistency, and trust are paramount. The systems I developed not only improved operational efficiency but also helped lay the groundwork for a smarter, more responsible AI hiring solution.

Moving forward, these contributions will continue to support ongoing model fine-tuning efforts, real-world deployments of Tara, and future enhancements to data quality and system scalability. This experience has been instrumental in strengthening my foundation in data engineering, machine learning, and prompt-based AI systems — and has left me with the confidence to pursue advanced challenges in AI product development.